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ARGUS: Learning a Bird from Five Calls

Two-Stage Few-Shot Detection of a Threatened Western Ghats Endemic, and a Measurement Gap in How the Field Scores It

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Per the IRIS anonymity rule, no school, city or state name appears anywhere in this document, its figures, or the accompanying video.

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Abstract

The Western Ghats is one of Earth's eight hottest biodiversity hotspots and one of its least acoustically monitored. *Sholicola albiventris*, a Western Ghats-endemic bird now IUCN Near Threatened, has only 22 labelled recordings in the largest public bioacoustic dataset available, short of what deep learning needs. Most species conservationists most need to track are data-poor for the same reason they are threatened. This project asks whether a model can learn to detect a new species from five example calls, and, harder, whether it can tell that species apart from ones that sound almost the same, not just detect a bird.

We built a two-stage pipeline. A small convolutional network, trained from scratch with a prototypical-network few-shot objective and fine-tuned in seconds per species, proposes candidate calls; a second stage embeds each candidate with Google's Perch v2 model and classifies it with a lightweight verifier trained on the same five examples. Across 10 Western Ghats species against their four hardest acoustic confusers, matching alone discriminated species barely above chance, mean AUC 0.585. Verification lifted all 10, mean AUC 0.751, a gain 5.2 times the measured noise floor, replicated independently four times, and cut false alarms on target-absent audio fourfold. Three further tests ruled out confounds: that this is just Perch, that the model learned recording equipment rather than the bird, and a stage-1 design choice.

Because it needs only five calls per species, this pipeline offers conservationists a practical route to monitoring threatened species too data-poor for conventional detectors to cover.

1. Introduction

1.1 The monitoring gap

The Western Ghats mountain range, running down India's western coast, is one of the eight most biologically rich hotspots on Earth, and among the least acoustically monitored. Passive acoustic monitoring — leaving recorders in the forest and analysing what they capture — is one of the few survey methods that scales to terrain where human surveyors cannot go often or quietly enough. Its bottleneck is not the recording. It is the listening: a single recorder generates more audio in a week than a specialist can review in a month.

Automated detection is the obvious answer, and for common species it works. The difficulty is that the species conservation most urgently needs to track are, almost by definition, the ones with the least data. *Sholicola albiventris* (White-bellied Sholakili), endemic to the higher-altitude shola forests of the Western Ghats and assessed as **Near Threatened** on the IUCN Red List [22], appears in the BirdCLEF 2024 dataset with **22 recordings**. Most other species in this study have 500. A conventional supervised classifier trained on 22 examples is not a serious proposition — and that is not a hypothetical: a custom BirdNET detector built for this same species by the research group that studies it required **several thousand songs** for training (C. Arvind, pers. comm., 4 Sep 2026). That system and this one are not directly comparable — different architecture, different task definition, different operating requirements — but the training-set sizes are the point. The data that approach needs does not exist for most of the species that need it. The gap matters beyond one species: India-wide assessment finds long-term declines concentrated among habitat specialists and endemics [23] — the same group that is hardest to gather training data for.

1.2 Why few-shot learning is the relevant regime

This scarcity is not a quirk of one bird. It is the structural condition of the problem: rarity is what makes a species worth monitoring and simultaneously what denies the monitoring system its training data. A detection method that requires thousands of labelled calls per species is therefore unavailable precisely where it would matter most.

Few-shot learning addresses this directly — learning a new class from a handful of examples by transferring a general notion of similarity learned elsewhere, rather than learning that class from scratch. The DCASE challenge series has run a five-shot bioacoustic detection task annually since 2021 [1,2,3], organised by

researchers at Queen Mary University of London and the Max Planck Institute of Animal Behavior, so the task is well-defined and actively contested. Reported F-scores have progressed from 38.4% (2021) to 65.2% (2024) [3,4].

1.3 The question this project actually asks

There is a distinction inside "detection" that the standard framing collapses, and it is the centre of this work.

Detecting that a bird called and **identifying which bird called** are different problems with different difficulty. A system that flags every bird-shaped sound in a recording can score well on a detection benchmark while carrying almost no information about species identity. For a labelling aid this may be acceptable. For a conservation record — *this Near Threatened species was present at this location on this date* — a confident wrong-species detection is worse than a miss, because it enters a database as a false positive record and is difficult to retract.

The DCASE task, as formally scored, cannot separate these. It is single-class within a single file: given five example events, find more events of that same class in that same recording [1]. There is no point at which a system is asked to distinguish the target from an acoustically similar non-target species, and ambiguous events are excluded from scoring rather than penalised. Notably, the challenge's own expert annotators *observe* cross-species confusion in top systems' outputs — one reviewer notes a system "mislabelled a bird call for a chirpgrunt" [1] — but the metric has no way to score it.

This project's research question is therefore: can a five-shot detector distinguish its target species from the acoustically nearest non-target species, and does a two-stage propose-then-verify architecture measurably improve that ability?

1.4 Objectives and design criteria

As an engineering project, this work states measurable design criteria in advance rather than a hypothesis:

#	Design goal	Criterion
1	Learn a new target species from **five** labelled recordings, with no retraining of the base encoder	Support set fixed at $n = 5$
2	Discriminate the target from its **four acoustically nearest** confusable species	Species-discrimination AUC, target vs. decoy
3	Improve on proposal-only matching by a margin **larger than run-to-run noise**	Gain must exceed the empirically measured noise floor
4	Generalise beyond the single study species	Tested on 10 pre-registered species
5	Reduce false alarms on audio containing **no** target	False alarms per hour on target-absent soundscape

Criterion 3 is deliberately phrased against a measured floor rather than a fixed threshold. An early version of this project reported improvements that were later found to lie inside run-to-run variation; measuring that variation directly, and testing every claim against it, is a core methodological commitment of the work (Appendix A).

2. Related Work

2.1 The few-shot bioacoustic task as the field defines it

Nolasco et al. [1] is the field's reference treatment, defining the five-shot detection task and analysing the 2021 and 2022 challenge editions with expert error analysis. Their §4.1 identifies, as an open problem, precisely the weakness this project measures:

> "The formation of a single fixed prototype, by taking an average in a coordinate space, implies that the examples are in some sense all of one kind… There remains much opportunity for multiplicity and drift to be included into designs for the concept formation of FSED systems." [1]

Prototypical networks [14] — the architecture family of the official DCASE baseline and of this project's stage 1 — collapse the five support examples to a single averaged point. The cost of that assumption on species identity had not, to our knowledge, been measured.

The same paper reports that shot *selection* is a dominant failure mode, with multiple independent expert annotators noting that the first five events in a file are frequently unrepresentative of the class [1]. DCASE fixes shots to the first five by protocol; this project varies the strategy explicitly (§4.5).

Liang et al. [4], the DCASE 2024 baseline paper, contributes hard-negative mining (+5.31 F1) and transductive query-time adaptation (+11.58 F1 alone; 46.78 → 63.67 combined), along with a controlled front-end comparison (log-Mel 63.67 vs PCEN 59.97 F1, but PCEN best-in-class on precision at 68.00).

2.2 Pretrained bird-audio embeddings

Ghani, Denton, Kahl and Klinck [5] — authors from the teams behind both Perch and BirdNET — establish the transfer-learning protocol this project's stage 2 implements: pretrained embeddings feeding a lightweight classifier head, which they show "can be harnessed for few-shot transfer learning, enabling the learning of new classes from a limited quantity of training data" [5].

Perch 2.0 [15] supersedes the model used in earlier versions of this work: EfficientNet-B3 backbone, 1536-dimensional embeddings, 14,795 classes, BirdSet AUROC improving 0.839 → 0.908 over Perch 1.0 with no fine-tuning. Two of its properties bear directly on this design. First, it is explicitly optimised for linear probing, which is exactly what this project's stage-2 verifier is. Second, its own prototype-learning classifier uses four prototypes per class with an orthogonality loss — the field's answer to the multiplicity problem [1] flagged above, applied during pre-training rather than to the five-shot adaptation problem studied here.

2.3 The gap this work occupies

The specific failure mode measured here — a detector that fires correctly but cannot confirm *which* species — is a case of **open-set recognition**: separating known classes from "none of the above" rather than assuming every input belongs to a training class [19]. Framing the contribution this way locates it inside an existing literature rather than presenting it as unprecedented.

Two further points from the field are relevant as sanity checks on task difficulty. At DCASE 2024, four in-context-learning systems submitted by a well-resourced research laboratory scored 1.2%, 3.3%, 6.5% and 8.1% F, against a 41.6% prototypical baseline on the same evaluation set; classical template matching scored 14.9% and cross-correlation 0.7%. The task is genuinely hard, and the obvious simpler alternatives are not close.

2.4 Adjacent work reviewed and not adopted

Recorded here because ruling an approach out is a design decision, and because a reader should be able to see which alternatives were considered rather than overlooked.

A different embedding. Self-supervised pretraining on unlabelled audio [7] and general animal-vocalisation encoders [9], benchmarked across taxa in BEANS [8], are the main alternative to a supervised bird-specific embedding. They were not used: Perch v2 [15] is trained on the taxonomy this study's targets belong to, and transfers even to underwater taxa well outside it [16], which makes the supervised embedding the stronger starting point for a bird.

Refinements to the prototypical baseline. Segment-level metric learning [10, 11], channel-spatial attention [12], and mutual learning between prototypical and classification branches [13] all improve detection F-score on the DCASE task. None was adopted, because §4.3 shows detection F-score and species discrimination are

not the same quantity, and it is the second that this project measures. Whether these transfer to discrimination is untested and would be a reasonable extension.

Reducing the labelling burden a different way. Active learning for avian bioacoustics [20] attacks the annotation budget rather than the five-shot constraint, and is complementary rather than competing. The fullest expression of that approach is agile modeling [26], from the group behind Perch: frozen pretrained embeddings, indexed audio search over a precomputed embedding database, and a human-in-the-loop active learning cycle that builds a working recogniser for a novel problem "in under an hour", demonstrated on coral reef sounds, juvenile Hawaiian bird calls and Christmas Island bird occupancy. It shares this project's central architectural commitment — keep the base model frozen and put a light classifier on top — and differs on the constraint that defines this work: agile modeling assumes an expert is available to label iteratively, whereas the five-shot setting studied here assumes five examples and no further human input. Where an expert *is* available, agile modeling is the better tool, and its indexed-search component is a more promising route to the adjudication problem of §4.12 than thresholding is.

Open-set recognition. Open-set strategies for long-term field recordings [21] — score thresholding, distance-based thresholding, OpenMax — are the closest existing framing to the problem measured here, and are the natural next step for the operating-point question raised in §5.3.

3. Methodology

3.1 Data

All audio is drawn from the **BirdCLEF 2024** dataset [6], a public competition dataset built specifically around under-studied Western Ghats species with limited training data. Recordings are sourced from Xeno-canto and carry per-recording metadata including species label, contributor ("author") identity, and a quality rating.

Audio is processed at **22,050 Hz**, mono, with STFT window 1024, hop 256, and 128 mel bins. These settings are identical to the DCASE 2024 baseline front-end [4], chosen so that the pipeline is directly portable to that benchmark rather than locked to a bespoke preprocessing stack.

3.2 Target and decoy selection, pre-registered

Target species (10). Selected by a rule fixed before any results were examined: species with at least *n*_shot + 4 recordings, ranked by recording count, with the study species *S. albiventris* forced in first so it cannot be dropped if it performs poorly. The resulting list is `whbsho3, bcnher, barswa, bkwsti, blrwar1, comgre, comkin1, commoo3, comsan, eaywag1`.

Decoy species (4 per target). The four species whose Perch-embedding centroids are *closest* to the target's — that is, the hardest available confusers, selected by acoustic similarity rather than convenience. This is a deliberately adversarial test design; §4.7 measures how much the headline result depends on it.

Encoder exclusion. All ten target species are excluded from the stage-1 encoder's training bank, not merely the primary one. An assertion enforces this at runtime. An earlier version of the pipeline excluded only the study species, allowing other targets to leak into the training data of the model meant never to have seen them; this was found and fixed before any reported result (Appendix A).

3.3 Stage 1 — proposal

Stage 1 is a small convolutional encoder (four convolutional blocks with adaptive pooling) trained **from scratch**, using episodic prototypical-network training [14]. Across 1,000 episodes, six species are drawn at random from a 40-species bank (targets excluded), with four support and four query clips each; the network is optimised so that each query clip embeds nearer its own species' support mean than any other's. This teaches a general-purpose acoustic similarity space rather than memorised species identities.

At detection time, for one target species and one recording: the five support recordings are expanded to three time-shifted crops each, injected into background soundscape at varied signal-to-noise ratios, and a lightweight classification head is fine-tuned on the copied encoder for **250 steps**. The recording is then scanned in overlapping 1-second windows, and adjacent windows above a deliberately permissive floor (`CAND_FLOOR = 0.10`) are merged into candidate events. The permissiveness is intentional: stage 1's task is recall, not precision.

3.4 Stage 2 — verification

Each stage-1 candidate is embedded with **Perch v2** [15] (1536-d; embedding width probed at load time, with a hard failure if it disagrees with the configured version, so that a mis-attached model cannot silently produce plausible numbers from the wrong network). Audio is DC-removed and peak-normalised to 0.25 before embedding, matching the official wrapper — an omission in an earlier version of this pipeline, and a correction to previously published numbers rather than merely part of the version upgrade.

Embeddings are scored by a **logistic-regression** verifier fitted on the five support calls as positives against many other-species calls as negatives, regularised at `C = 0.1`. Strong regularisation is necessary rather than cosmetic: with roughly 100 training points in 1536 dimensions an unregularised model separates the classes perfectly, which destroys the score's ability to *rank* candidates — everything saturates at 0 or 1.

The pipeline emits, per candidate, both stage-1 confidence p and verifier confidence v. Six fusion rules combining them were evaluated (§4.1).

3.5 Evaluation protocol

Three evaluation passes run on the same trained detector, each answering a different question:

Pass	What it plants	What it measures
`eval_pass`	Real target calls in clean background	Detection quality; sets the operating threshold
`spec_pass`	Target calls **and** decoy-species calls	Species discrimination (the primary readout)
`neg_pass`	Nothing — clean soundscape, zero target calls	False alarms per hour (every detection is false by construction)

Held-out calls are strictly disjoint from support calls. Unless otherwise stated, calls are planted at +6 dB SNR; §4.6 varies this.

Seeds. The centrepiece uses three encoder-training seeds (0, 1, 2) per species, so that reported variation includes encoder-training variance rather than only detection-time randomness. The sweeps of §4.4, §4.5 and §4.7 use 5 species × 3 detection seeds — 15 measurements per condition — widened from an earlier 3 × 2 (6 per condition) specifically because six is below the level at which a difference of this size can be distinguished from noise. Where widening changed a conclusion, both samplings are reported.

3.6 Metrics, and why AUC leads

The primary metric is **species-discrimination AUC**: given one held-out target call and one decoy-species call at random, how often does the system score the target higher. 0.50 is chance; 1.00 is perfect. It is threshold-free, so it does not depend on a selected operating point.

This choice is defended rather than assumed, because a substantial literature argues ROC-AUC is optimistic under class imbalance and that precision-recall is preferable for rare events. Beam [18] gives the cleanest treatment: AUROC has greater resolving power in general, but AUPRC is better "when searching among high-quality classifiers applied to low prevalence outcomes." Neither condition holds here — the specificity test plants 6 targets against 12 decoy items (1:2, mildly imbalanced, not low-prevalence), and a 0.585 baseline classifier is not high-quality. Independently, BirdCLEF 2024 selected macro-averaged ROC-AUC as its official metric explicitly because "as a threshold-free metric, ROC-AUC allows comparing overall model quality,

without requiring participants to engage in difficult (and opaque) threshold-selection processes" [6], and the Perch 2.0 authors state that on BirdSet "we believe that the ROC-AUC score is the most stable and informative" [15].

Average precision (AP) and event **F1** are also computed and reported, because the divergence between them and AUC is itself a finding (§4.3). AP is the better summary for the *detection* task, which is genuinely low-prevalence; AUC is the right one for *discrimination*. Reporting both, and naming which question each answers, is the position this paper takes.

A comparability caveat stated in advance: BirdCLEF's ROC-AUC is macro-averaged over 108 species on soundscapes with full training data. This project's AUC is target-vs-decoy discrimination from five examples. These are not the same quantity and are never compared numerically in this paper.

3.7 The measured noise floor

Because GPU floating-point operations are non-deterministic in ordering, identical configurations do not produce identical results. The magnitude of that variation is measured directly, not assumed: the baseline configuration was run across three encoder seeds on three species, giving per-species spreads of 0.035, 0.029 and 0.032 verified AUC — a **mean noise floor of 0.032, maximum 0.035** (`data/argus_ablation_ladder.csv`, `config == "pcen|logreg"`).

Every comparative claim in §4 is tested against this floor. Where a difference does not clear it, this paper reports the result as unresolved rather than as a finding.

3.8 Reproducibility and verification practice

The pipeline is five Python files executed as ordered notebook cells. The notebook is *generated* from those files with a SHA-256 equality check rather than hand-copied, so the executed code is verifiably the tested code. **269 local assertions** exercise the pipeline's pure logic — scoring mathematics, verdict rules, cache-key correctness, structural checks on every skip flag — without requiring a GPU, Perch, or the dataset.

Every aggregate reported in this paper was independently recomputed from the raw CSVs during writing, and disagreements between recomputation and previously published summaries were resolved in favour of the raw data (Appendix B).

4. Results

All figures below are computed from the CSV files in `data/`. Regeneration commands are in Appendix B.

4.1 Primary result: verification generalises across all ten species

Across 10 species × 3 encoder seeds:

	Stage 1 alone (matching)	+ Verification (Perch)
Mean species-discrimination AUC	**0.585**	**0.751**
Mean gain	—	**+0.165**
Species improved	—	**10 / 10**
Gain ÷ standard error	—	**8.83**
Gain ÷ measured noise floor (0.032)	—	**5.2×**

Per species, sorted by gain:

Species	Stage 1	Verified	Gain
comgre	0.687	0.781	+0.094
comsan	0.623	0.720	+0.097
blrwar1	0.535	0.646	+0.111

Species	Stage 1	Verified	Gain
commoo3	0.628	0.750	+0.122
bkwsti	0.694	0.828	+0.134
comkin1	0.621	0.827	+0.205
eaywag1	0.545	0.753	+0.208
whbsho3 (study species)	**0.469**	**0.680**	**+0.211**
barswa	0.554	0.770	+0.216
bcnher	0.496	0.750	+0.254

Two observations. First, the improvement is universal — every species improves, and the smallest gain (+0.094) still exceeds the 0.032 noise floor by roughly 3×. Second, stage 1 alone is at or below chance for two species (whbsho3 0.469, bcnher 0.496): for these, proposal-only matching carries *no* usable species information, and the entire discriminative signal comes from verification.

Fusion rules. Six ways of combining stage-1 confidence *p* and verifier confidence *v* were compared:

Fusion rule	AUC	AP
v alone (verifier only)	**0.751**	0.604
sqrt(p) · v	0.650	0.636
p · v	0.626	0.625
sqrt(p · v)	0.626	0.625
p · sqrt(v)	0.609	0.619
p alone (stage 1 only)	0.585	0.611

Using the verifier's confidence *alone* — discarding stage 1's confidence entirely — outperforms every blend. This was not the expected outcome, and it clarifies the division of labour: stage-1 confidence reports how strongly something bird-like is present, which is useful for *locating* candidates but adds essentially no species information once the verifier has assessed them. Blending mostly injects noise into a question already better answered.

Replication. The centrepiece has been run in four independent full sessions. Mean gains across sessions: +0.165, +0.166, +0.161, +0.165. The largest single per-species movement between the two most recent sessions was 0.013.

4.2 False alarms on target-absent audio

On soundscape recordings containing no known target call, at the best-F1 operating threshold:

	Stage 1 alone	Verified
Mean false alarms / hour	**54.0**	**13.4**
Reduction	—	**4.03×**
Worst species	78.8 (whbsho3)	54.2 (comkin1)
Best species	23.0 (bkwsti)	0.7 (barswa)

Two caveats are stated here rather than left for a reader to find. First, these beds are real Western Ghats soundscapes, not silence — any genuinely wild target call present in them is counted against the system, so every figure is an upper bound on the true false-alarm rate. Second, a recall-matched control at 90% recall is **not** usable at this operating range: peak recall anywhere in the run is 0.778, so a 90%-recall threshold never binds and the two arms become statistically indistinguishable (237.9 vs 236.9/hr). The 4× reduction is measured at the best-F1 threshold, and that is the only false-alarm claim these data support.

4.3 The measurement gap: detection retrieval versus species discrimination

Comparing soundscape event retrieval performance with targeted species discrimination against acoustic confusers:

Metric	Stage 1	Verified	Change	Species improved
Soundscape retrieval AP	0.611	0.604	**−0.008**	**4 / 10**
Event F1	0.675	0.710	+0.035	9 / 10
Species-discrimination AUC	0.585	**0.751**	**+0.165**	**10 / 10**

While soundscape retrieval average precision slightly declines after verification (0.611 → 0.604), species-discrimination ROC-AUC against near acoustic confusers rises substantially (+0.165 across all ten species). The divergence is prominent when ranking the six candidate fusion rules: ranking by soundscape retrieval AP places the verifier-alone pipeline last (0.604), whereas ranking by species-discrimination AUC places it first (0.751).

This divergence stems from distinct evaluation objectives. Soundscape AP evaluates event retrieval—whether acoustic calls are detected against background noise across variable confidence thresholds. However, it does not explicitly isolate fine-grained discrimination between target calls and acoustically similar non-target species. In contrast, the species-discrimination AUC evaluates pairwise ranking between the target and its four hardest acoustic confusers. Consequently, an evaluation focused exclusively on soundscape call retrieval would fail to recognize the configuration with the strongest taxonomic discrimination.

4.4 How many examples are actually needed

The project's premise is five examples. That number was tested rather than assumed (verified AUC, 5 species × 3 seeds — 15 measurements per condition):

Shots	Stage 1	Verified	Step
1	0.568	0.680	—
3	0.555	0.720	**+0.040**
5	0.543	0.726	+0.007
10	0.565	0.758	+0.032

The curve rises monotonically, with **+0.078 verified AUC from 1 to 10 shots** — about 2.4× the 0.032 noise floor. But only the **1 → 3** step clears that floor on its own (+0.040). Going from 3 to 5 buys +0.007, comfortably inside noise, and 5 to 10 lands at +0.032, exactly on it.

The honest reading is that more examples always help, that the return is concentrated in the first few, and that "five" is a defensible operating point on a smooth curve rather than a sharply justified optimum. An earlier version of this project claimed performance "plateaus at five shots"; that was stronger than the data supported and was retracted (Appendix A). Widening from 3 species × 2 seeds to 5 × 3 did not rescue the retracted claim — it confirmed the correction, with the 3 → 5 step shrinking further from +0.031 to +0.007.

Note also that stage 1 alone stays in a narrow band (0.543–0.568) at *every* shot count, with no upward trend. Additional examples do not rescue proposal-only species discrimination; they improve the verifier.

4.5 Which five examples are chosen

Four shot-selection strategies were compared: highest audio-quality rating (`rated`), the first five in the dataset (`first`, matching DCASE protocol), a random draw, and a `diverse` strategy maximising pairwise embedding distance (5 species × 3 seeds):

Strategy	Stage 1	Verified
random	0.556	0.748

Strategy	Stage 1	Verified
rated	0.542	0.735
diverse	0.541	0.731
first	0.544	0.715

The spread across strategies is **0.033** — indistinguishable from the 0.032 noise floor. At the narrower 3 × 2 sampling this spread was 0.068, and an earlier version of this project read it as a real effect favouring `rated`; that claim was retracted (Appendix A) after a two-way test returned F(3,6) = 0.54, below what noise alone produces. Widening to 5 species × 3 seeds halved the spread and moved the nominal winner again, from `rated` to `random` — the third different aggregate winner across runs. Both facts point the same way: **no strategy reliably wins in aggregate.**

Per species the picture is not flat — `diverse` was best for the study species in 4 of 4 earlier runs and worst for a different species in 4 of 4. Shot choice matters; it does not have one universal right answer.

That `first` ranks last in both samplings is worth noting, since it is the strategy the DCASE protocol mandates, and it aligns with the expert annotator observations reported in [1]. It is a consistent ordering rather than a significant one.

4.6 Robustness to call loudness

Every other result in this paper is measured at a fixed +6 dB SNR — loud, close calls. This experiment varies planted-call loudness from +9 dB down to −6 dB (5 species × 3 seeds, 15 measurements per level). **The pre-registered prediction was that the verifier's advantage would shrink as calls became fainter.** It did not, and the reason requires distinguishing two quantities that are easy to conflate:

SNR (dB)	Verified AUC, *all* planted	Verified AUC, *localised only*	Target coverage
+9	0.737	0.848	0.813
+6	0.738	0.839	0.820
+3	0.727	0.841	0.804
0	0.714	0.841	0.784
−3	0.675	0.839	0.717
−6	0.634	0.820	0.647

Tested per species and paired across the +9 → −6 range (*n* = 5 species, seed-averaged):

Quantity	Change	d/SE	Species consistent
Target coverage	**−0.167**	−5.35	5/5
Verified AUC, all planted	**−0.103**	−8.71	5/5
Verified AUC, localised only	−0.028	**−1.34**	4/5

The interpretation: **what degrades in quiet audio is detection, not identification.** Coverage falls by a sixth and the all-items AUC falls with it, both unambiguously. But over calls the system actually *localised*, discrimination is statistically indistinguishable from flat — and across +9 dB to −3 dB it varies by only 0.009, with the entire spread produced by the −6 dB point alone. The apparent decline in the all-items figure is driven by calls never localised, which are scored 0.0 by construction.

Reporting coverage alongside the AUC is therefore not optional. An AUC drifting toward 0.50 at low SNR is otherwise ambiguous between "target and decoy became indistinguishable" and "almost nothing was detected, so nearly everything tied" — two entirely different findings.

A disagreement between tests, disclosed rather than resolved in our favour. Stage 1's *own* localised discrimination changes by −0.088 across the same range (d/SE −4.46, negative in 5 of 5 species). An earlier version of this project claimed exactly this — that stage-1 specificity degrades with SNR — and **retracted it** (Appendix A, #39) after an audit found it had been tested against the wrong noise series. Under the margin-

versus-bound test this project uses for its sweeps, the widened data still returns *too close to call*, because that test bounds against the worst single cell's standard deviation (0.141) and the change is smaller than that. Under the paired per-species test used for the centrepiece, the same data is clearly significant.

Both tests are used elsewhere in this paper, for different purposes, so neither is being introduced for the occasion — but they disagree here, and the more favourable one is not adopted on that basis. The retraction stands. What can be said is narrower and is said instead: the widened data is *directionally consistent* with the retracted claim, and a properly powered test designed in advance would be needed to revisit it. Reversing a retraction because a second test was more generous is the precise failure the retraction was issued to correct.

4.7 Were the decoys chosen to flatter the result?

Decoys have always been the hardest available confusers by design. This experiment measures how much the headline depends on that (5 species × 3 seeds):

Decoy tier	Stage 1	Verified	Verification gain
hard (closest confusers)	0.546	0.726	**+0.180**
medium (5th–8th closest)	0.598	0.760	+0.162
random (whole pool)	0.607	0.782	+0.176

Absolute AUC rises as decoys get easier, exactly as it must by construction — that ordering is a sanity check on the decoy ranking, not a finding.

The question that matters is whether verification's *gain* depends on difficulty, and it does not: +0.180 / +0.162 / +0.176, a spread of **0.018**, comfortably inside the 0.032 noise floor and with no monotonic trend. At the narrower 3 × 2 sampling this spread was 0.037, marginally above the floor; widening it halved the spread and moved the result from "barely flat" to "clearly flat."

The direction matters for honesty: easier decoys produce a **higher** headline number (+0.056 verified AUC for random over hard). The reported figure is the conservative choice, not a flattering one.

4.8 Ablation: four published techniques, all rejected

Four binary design choices were tested factorially (8 configurations, 3 species): verifier type (logistic regression vs. prototypical probe), hard-negative mining, and front-end (PCEN vs. log-Mel). Each is a technique with published support — the prototypical probe from Perch 2.0's own benchmark [15] (+0.078 mAP on detection tasks), hard-negative mining from Liang et al. [4] (+5.31 F1), log-Mel from both the DCASE 2024 winning entry (65.2% vs 61.5% for the same team's PCEN build) and Liang et al. Table III.

Configuration	Verified AUC	False alarms / hr
pcen | logreg (baseline)	**0.742**	**8.2**
pcen | proto	0.724	27.8
pcen | logreg + hard-neg	0.676	12.3
pcen | proto + hard-neg	0.667	37.3
logmel | logreg + hard-neg	0.645	8.0
logmel | logreg	0.605	3.3
logmel | proto + hard-neg	0.598	28.8
logmel | proto	0.588	37.0

The untouched baseline beat every alternative. A fifth technique, TimeFilterAug — the augmentation from the DCASE 2023 winning entry, described in [17] — measurably degraded every metric in earlier testing and was dropped from the factorial design.

The rejection of the prototypical probe and hard-negative mining rests primarily on **false-alarm rate**, which rose 3× to 19× across independent runs — a large, reproducible cost. Their effect on raw AUC was originally

reported as larger than it is; that framing has been corrected (Appendix A), but the false-alarm-based rejection is unchanged.

Log-Mel is the interesting case, and it repays a closer look. Both published sources predict it should *outperform* PCEN by ~3.7 F-points, and here it does the opposite on AUC by a wide margin (0.605 vs 0.742) — while apparently delivering the lowest false-alarm rate in the table (3.3/hr). That combination looks like a precision-for-recall trade. It is not. Paired within matched (species, verifier, hard-negative) cells, $n = 12$:

Metric (verified)	log-Mel − PCEN
Recall	**−0.220**
Precision	−0.119
F1	−0.206
AUC	−0.092
False alarms / hr	−2.6
Recall (stage 1)	**−0.192**

Verified recall falls from 0.610 to 0.366 — the log-Mel pipeline simply detects far less. Its lower false-alarm count is the arithmetic consequence of firing less often, not of discriminating better; precision falls alongside recall rather than rising to meet it. The collapse originates at stage 1, whose own recall drops 0.570 → 0.359, and propagates: stage 2 cannot verify candidates that were never proposed. Notably, stage-1 *AUC* rises slightly under log-Mel (+0.027) even as its recall collapses, so the front-end ranks what it does find marginally better while finding much less of it.

This reconciles the divergence from published expectation rather than leaving it open. The reported log-Mel advantage [4] was measured on fully-trained systems with substantial task-specific training; here the stage-1 encoder is fine-tuned for 250 steps from five examples. PCEN's per-channel gain control appears to matter considerably more in that short-adaptation regime than in the regime where the published comparison was made. The claim this supports is narrow and stated as such: log-Mel is worse **in this five-shot, short-fine-tune setting**, by a mechanism (recall collapse at the proposal stage) that the published comparisons had no occasion to observe.

That four techniques with published support all failed to transfer to this five-shot, short-fine-tune regime is itself a result, and is reported rather than omitted.

4.9 Three confounds, closed

An adversarial review identified three alternative explanations the pipeline had never tested. All three were run (5 species, 2 seeds, 3.34 hours on one GPU session; `data/argus_confounds.csv`, 55 rows).

P1 — "Isn't this just Perch?" — settled. Perch was run as a standalone detector, scanning recordings directly with no stage-1 CNN in the path at all.

Arm	Species AUC	Mean recall
Perch alone	**0.526**	**2.5%**
Stage 1 alone	0.536	—
Verified pipeline	**0.735**	—

Perch alone collapses to near chance and finds almost nothing — 2.5% recall. The verified pipeline exceeds it by **0.209**, far clear of the noise bound. A plausible mechanism: Perch's 5-second analysis window is not sharply localised in time, so its peak response can fall outside the ±0.7 s tolerance a 1-second call requires. Neither stage substitutes for the other; the architecture is not decorative.

P2 — does stage 1's negative-sampling choice drive the result? — settled, and it reverses the literature's expectation. Stage-1 fine-tuning has always drawn its negatives from arbitrary background crops. Replacing them with species-matched negatives — real calls of other species — is what both Nolasco et al. [1] (negative choice named as decisive) and Liang et al. [4] (+5.31 F1 from hard-negative mining) would predict should help.

At two seeds this was genuinely too close to call in both directions, and was reported as such. It was re-run at **six seeds** (5 species, 20 measurements per arm):

Arm	Background negatives	Species-matched	Change	Bound	Verdict
Stage 1 alone	0.550	0.583	+0.034	0.046	**flat**
Verified	0.737	0.686	**−0.050**	0.040	**falls**

Species-matched negatives **hurt** the verified pipeline, and the stage-1 improvement that looked promising at two seeds (positive in 5 of 5 species) turns out to sit inside noise once properly powered.

Two honest qualifications. First, the falls verdict clears its threshold narrowly — margin 0.0105 against a 0.0099 knife-edge. What actually carries the result is per-species consistency: the change is negative in **5 of 5 species**, which would occur by chance with probability 0.03 if direction were arbitrary. The margin test alone would be thin evidence; the consistency is what makes it reportable. Second, the substitution is not clean — a 240 s bed cannot physically hold 200 non-overlapping 1 s calls at 0.4 s spacing, so the species-matched arm is approximately 126 species-matched plus 74 background negatives, not a pure swap. The comparison is "mostly species-matched versus all background," and is stated that way rather than as a clean factor.

This result does not stand alone. The ablation ladder (§4.8) independently found hard-negative mining degraded the pipeline. Two separate experiments, different mechanisms, same direction: **in this five-shot, short-fine-tune regime, making stage 1's negatives harder makes the pipeline worse.** A plausible reading is that a small, brief adaptation budget is better spent learning what the target sounds like than learning fine distinctions against a narrow set of confusers — but that is an interpretation these data suggest rather than test.

P3 — did the model learn the recording setup rather than the bird? — settled. The held-out pool was rebuilt so it can never share a contributor ("recordist") with the five support calls, removing any possibility that shared recording equipment, location or technique explains the discrimination.

Arm	Stage 1	Verified
Shared recordists permitted	0.531	0.740
Recordist-disjoint	0.531	0.708
Change	−0.000	**−0.032**

The verified change of −0.032 sits against a 0.062 bound: flat. One species (bcnher) moved further than the rest (−0.121, still inside its own wider bound) and is disclosed rather than smoothed over. Recording-setup leakage is not what the specificity result is measuring.

4.10 Distilling Perch into stage 1: what deployability costs

Every result above uses Perch v2 (~12M parameters, 14,795 training classes) as the verifier's embedding source. That is a legitimate use of a published transfer-learning protocol [5], but it means the part of this pipeline that actually discriminates species is not something this project built. It is also a genuine deployment constraint independent of that question: Perch cannot run on a battery- or solar-powered field recorder, while a network of a few hundred thousand parameters can.

Both concerns point at the same experiment: distil Perch's embeddings into the same stage-1 network already used for candidate proposal, and ask what discrimination survives once **no Perch inference runs at detection time at all**.

Method. A linear head (128 → 1536) was added to the unmodified stage-1 trunk and trained by cosine loss to reproduce Perch's embedding of the same one-second audio crop, using 2,636 (spectrogram, waveform) pairs from 60 non-target species. The validation split was held out by **species**, not by crop — 9 of 60 species never appeared in training — so the reported similarity measures transfer to unseen species rather than memorisation of training recordings. Best held-out-species cosine similarity to Perch: **0.399** after 60 epochs.

Everything downstream — decoy ranking, shot selection, stage-1 candidate proposal, the verifier's fitting procedure — was held identical between the two arms; only the embedding space the verifier is fit and scored in was swapped.

Deployed size. The trunk alone is the ~445k-parameter network stage 1 already runs. Matching Perch's output space requires the projection head as well, a further ~198k parameters, for a deployed total of **642,688 — about 19× smaller than Perch**, not the 27× an earlier estimate of trunk-only size implied. That correction is made here rather than left standing, consistent with this project's practice elsewhere.

Result. On the same 5-species specificity test used throughout §4.9 (2 seeds):

	Perch verifier	Distilled verifier	Change
Mean species AUC	0.733	0.596	**−0.136**
Mean false alarms / hr	10.2	84.2	**8.3× worse**

The change is 5.4× the 0.025 noise bound — unambiguous, not a close call. The pre-registered prediction (Appendix, `arguswala_distill.py`) named three possible outcomes before this ran: a loss small enough to be free, a large but bounded loss describing a real accuracy/deployability frontier, or collapse to chance describing a negative result. **The honest answer is not one of the three cleanly — it is per-species heterogeneous, straddling the second and third:**

Species	Perch AUC	Distilled AUC	Change
bkwsti	0.814	0.803	**−0.011**
barswa	0.789	0.718	−0.071
blrwar1	0.594	0.537	−0.056
whbsho3 (study species)	0.672	**0.511**	**−0.161**
bcnher	0.795	**0.412**	**−0.383**

For bkwsti, compression is very nearly free — within the noise floor. For barswa and blrwar1, it is a real but moderate cost, consistent with a genuine accuracy/deployability frontier. For bcnher and, more consequentially, for ***S. albiventris*** **itself**, the distilled verifier lands at or below chance (0.50) — a collapse, not a frontier. The species this project is framed around is one of the two that fails.

There is no clean predictor of which species survive distillation in this data — the teacher's own AUC does not order the losses (bcnher scores second-highest under Perch and loses the most; blrwar1 scores lowest under Perch and loses comparatively little). The false-alarm increase is not uniform either: it is concentrated in the same two collapsing species (whbsho3 197/hr, bcnher 22/hr) rather than spread evenly.

What this does and does not establish. It does not support a claim that this pipeline can be deployed today without Perch — for the species that motivates the project, it demonstrably cannot yet. It does establish, with a real measured number rather than an assumption, that a ~445k-parameter network trained from scratch on 40 species carries substantially less of Perch's discriminative structure than a linear probe on Perch's own embeddings does — the gap the rest of this paper's architecture exists to bridge. Whether that gap narrows with more distillation data, more training species, or a non-linear projection head is future work (§7), not a question this experiment was scaled to answer. Asked directly, Perch's own author judged the heterogeneity unsurprising and named two candidate causes: gaps or imbalance in the student's per-species training data, so that rarer species degrade first, and insufficient capacity — specifically layer width, which the same group found to matter when ablating a pretrained model's own embedding dimension (Fig. 4 of [5]: Perch at 320-d is "significantly degraded ... in all tasks", and doubling to 2560-d improves some) (T. Denton, pers. comm., 4 Sep 2026). The width sweep was already running when that reply arrived; it therefore tested a mechanism the model's authors independently nominate rather than one this project guessed at, and §4.11 reports what it found. Per-species training-set balance is a second, untested explanation and is not distinguishable from width in the present data; 2,636 training pairs against a 1536-dimensional target is a small fit, and the honest

reading is that this result bounds what a modest distillation effort achieves, not what distillation can achieve in the limit.

4.11 Is the bottleneck the constraint? A width sweep

§4.10 left one specific mechanical explanation for the distillation loss untested. Perch emits 1536 dimensions; the student's trunk emits 128. A linear head from a 128-dimensional trunk can only reproduce a 128-dimensional subspace of Perch's space, however long it trains. If that is the binding constraint, the held-out cosine of 0.399 is a ceiling rather than a symptom, and widening the trunk should lift both fidelity and discrimination together.

Three outcomes were pre-registered in `arguswala_frontier.py` before the run: both rise (the bottleneck was binding), both stay flat (it was not, and the cause is data or optimisation), or **fidelity rises while species AUC does not** — which would mean the student reproduces Perch's embedding geometry in aggregate while losing the part that separates confusable species.

Method. One teacher set of 10,163 (spectrogram, waveform) pairs from 98 non-target species — 3.9× §4.10's — built once and shared by every arm, split 8,527 / 1,636 by species (83 training species, 15 held out). Bottleneck widths 64, 256 and 512 alongside §4.10's 128, plus one nonlinear-head arm at 512 to separate "too few dimensions" from "a single linear map is not enough". Identical stage-1 encoder, identical train/held-out partition, 90 epochs each, and a same-session Perch control rather than a carried-over number. Evaluation is the standard 5-species × 2-seed specificity test used throughout §4.9.

Arm	Parameters	vs Perch	Held-out cosine	Species AUC	FA/hr
Perch v2 (control)	~12M	—	—	**0.727**	8.7
distil-64	470,528	25.5× smaller	0.4446	0.610	49.6
distil-128	642,688	18.7× smaller	0.4735	0.617	51.6
distil-256	987,008	12.2× smaller	0.4998	0.617	36.6
distil-512	1,675,648	7.2× smaller	0.5226	0.611	48.0
distil-512 + MLP head	1,938,304	6.2× smaller	0.5042	0.633	46.6

Result: the third outcome, unambiguously. Across an eightfold widening, fidelity to the teacher rises monotonically — 0.4446 → 0.4735 → 0.4998 → 0.5226, a gain of 0.078 — while species discrimination does not move: 0.610 → 0.611, and the total spread across all four linear widths is **0.0073, well inside the measured 0.032 noise floor (§3.7)**. This is flat, not weakly positive.

The bottleneck is therefore **not** the binding constraint, and the interpretation is the one the pre-registration flagged as most interesting: a student can be trained to sit progressively closer to Perch's embedding geometry and gain nothing whatsoever at telling two similar birds apart. Whatever carries species identity in Perch's representation is not what cosine similarity to the teacher is measuring. That is a statement about distillation objectives, not about this particular student — and it is a caution to anyone compressing a bioacoustic embedding model by matching embeddings and validating on embedding similarity.

The nonlinear arm is not a result. At 512-d the MLP head scores 0.633 against the linear head's 0.611. The difference is +0.021, *inside* the 0.032 noise floor, so this paper does not claim that a nonlinear map helps. It is reported because it was pre-registered, and because reporting only the arms that cleared threshold would be selective.

What did move: the teacher set. At the identical 128-d linear configuration §4.10 published, *S. albiventris* scored 0.511 — chance — on a 2,636-pair teacher set. Here, on 10,163 pairs, it scores **0.608**: a gain of 0.097, three times the noise floor, at the same width and the same head. This is a between-session comparison and not a controlled one — epochs (60 → 90) and batch size (64 → 128) changed alongside the data — so it is suggestive rather than settled. But it points at scale of the distillation set, not at architecture.

Asked before this ran, Perch's author named two candidate causes (§4.10): per-species gaps or imbalance in the student's training data, and insufficient capacity, specifically layer width. On these data, **width does not explain the loss; data plausibly does.**

It would be too glib to call that a failed prediction, and checking the cited evidence shows why. Figure 4 of [5] ablates the **embedding dimension of a pretrained model** — one trained end-to-end on a large supervised classification objective — and finds it matters a great deal: Perch at 320-d is "significantly degraded ... in all tasks", and doubling to 2560-d improves some. That is a different experiment from the one here, which varies the bottleneck of a small student regressed onto a *fixed* 1536-d target. Width being decisive in the first setting and inert in the second is not a contradiction.

It is, though, informative about which of the two settings this project is in. Widening did buy what that work would predict — steadily better fidelity to the teacher. What it did not buy was any of the thing the fidelity was supposed to be a proxy for. When more capacity improves the training objective and leaves the downstream measurement untouched, the constraint is not the capacity. It is the objective.

4.12 First scan of unlabelled wild audio

Every number to this point comes from target calls *planted* into real soundscape (§3.5). The pipeline had never been pointed at continuous unlabelled audio and asked what is in it. This section reports the first such scan and is deliberately cautious about what it can support.

Method. 400 soundscape recordings from the same dataset, strictly disjoint from the 14 used for calibration and evaluation — 26.6 hours of audio, scanned in 91 minutes. Both the study species and a control species (*Himantopus himantopus*, a target with an entirely different niche) were scanned inside a single pass over the files, so their file sets are identical by construction. Each ran at its own best-F1 operating threshold derived from planted evaluation, never tuned on these files, which have no labels to tune against.

	Detections	Files	Rate	Threshold	Mean score	Max score
S. albiventris	**350**	55	**13.2 / h**	0.4909	0.4981	0.5264
Control (*H. himantopus*)	201	66	7.6 / h	0.4726	0.4774	0.5073

What this establishes. On identical audio, the study species fires at **1.74× the control rate**. That is the first evidence in this project that the pipeline behaves differently on wild audio depending on which species it has been given five examples of, rather than simply firing on bird-shaped sound.

What it does not establish, stated plainly.

1. **No detection here is confirmed.** There is no ground truth for these recordings, and nobody on this project can identify *S. albiventris* by ear. The 60 highest-scoring detections per species were saved as audio clips for adjudication by someone who can; that has not happened.
2. **Every score sits marginally above its threshold.** The study species' highest-scoring detection in 26.6 hours scores 0.526 against a 0.491 threshold — 0.035 of headroom — and the mean clears by 0.007. Nothing fires confidently. A detector genuinely finding its target in the wild would be expected to produce at least some high-confidence hits, and this did not.
3. **The two thresholds are not identical** (0.491 vs 0.473), each being that species' own best-F1 point, so the rate ratio bundles a small threshold difference with any real effect.
4. **The control was detected in *more* files** (66 vs 55) despite fewer total detections, so the study species' hits cluster within fewer recordings — consistent either with a real territorial bird or with a handful of recordings containing a persistent confusing sound.

What the shortlist itself looks like. The 120 saved clips (2.0-7.4 s, mean 2.7 s) carry their source recording in the filename, which allows two checks before anyone listens to them. First, the top detections are concentrated: the study species' 60 highest-scoring detections come from just 22 distinct soundscapes of the

400 scanned, and its busiest single recording contributes 10 of them. The control is similarly concentrated (25 files, busiest contributing 16). Concentration of this kind is consistent with a territorial bird recorded repeatedly in one place, and equally consistent with one recording containing a persistent confusing sound; the clips cannot distinguish those and neither can this paper.

Second, and more usefully, the two species' top-60 lists share only **three** source recordings. Under independence one would expect about 1.4 (22 x 25 / 400), so the observed overlap is roughly twice chance — which on counts this small distinguishes nothing, but does rule out the worst case. Had the two lists largely coincided, it would have been direct evidence that the pipeline was surfacing the same generically detector-attracting audio regardless of which five calls it was given. They do not, and those three shared recordings are the first files an expert should be asked about.

The honest summary is that this produces a **rate, a ranked shortlist, and a comparison** — which is what was pre-registered — and that the shortlist now needs an expert ear before any of it can be called a detection. It moves limitation 2 from "never attempted" to "attempted, unadjudicated", which is progress but not resolution.

5. Discussion

5.1 What the two stages each contribute

The results support a clean division of labour that was hypothesised architecturally and is now measured directly.

Stage 1 is a **locator**. It is fast, recall-oriented, and — as §4.1 and §4.4 both show — carries close to zero species information on its own, sitting near chance regardless of how many support examples it is given. Stage 2 is an **identifier**. It is accurate on species but, as P1 demonstrates, nearly useless at finding events on its own, recovering 2.5% of calls when asked to scan directly.

Each stage fails at the other's task, and the failure is measured rather than asserted. This is a direct answer to two obvious challenges: "your CNN isn't doing anything, it's all Perch" (P1: Perch alone reaches 0.526) and "then why not just use the CNN" (§4.1: stage 1 alone reaches 0.585, below chance for two species).

A sharper version of the first challenge remains, and §4.10 was built specifically to answer it: not "does the CNN contribute nothing," but "is the part that actually discriminates species yours." The honest answer, after distillation, is *partially, and unevenly*. The stage-1 network can be extended to carry meaningful species discrimination without Perch at detection time for some species (bkwsti loses 0.011 AUC; barswa and blrwar1 lose a moderate, bounded amount) but not reliably for others — most consequentially not yet for *S. albiventris* itself, which collapses to 0.511, indistinguishable from chance. This does not overturn the division of labour above; it quantifies how much of stage 2's contribution is currently irreducible to a ~640k-parameter model trained the way this project trains one, which is a more precise and more defensible claim than either "the CNN does nothing" or "the CNN could do it all with more effort."

5.2 Why the metric gap matters beyond this project

§4.3 demonstrates that standard bioacoustic detection metrics and species discrimination metrics capture different phenomena. Evaluating call retrieval alone provides little visibility into whether a detector reliably separates target calls from acoustically similar confusers. As shown in Table 4, a configuration optimized purely for soundscape event retrieval can rank poorly on targeted species discrimination against near decoys.

This is consistent with, and extends, what the field's own reference paper already documents: expert annotators observe cross-species confusion in top systems' outputs, and standard retrieval scoring protocols have limited mechanisms to isolate and penalise it [1]. The contribution here is to quantify that divergence on real data and demonstrate that prioritizing soundscape retrieval alone can obscure genuine species discrimination gains.

The practical recommendation follows directly: for few-shot bioacoustic detection intended to produce **species records** rather than general labelling candidates, a discrimination metric against acoustically near non-targets should be reported alongside general retrieval metrics. Neither replaces the other.

5.3 Two operating points, two use cases

There is a tension in this work's framing that the field's own expert feedback raises, and it is better volunteered than discovered. A coati expert in [1] states:

> "In general, I prefer over-predicting calls to under-predicting, as deleting incorrect labels is faster than listening to whole wave files." [1]

By that standard, optimising to *reduce* false alarms (§4.2) is the wrong direction. The resolution is that these are two different deployment modes with genuinely opposite requirements:

- **As a labelling aid** — surfacing candidate events for a human to review — recall dominates, over-prediction is cheap, and the expert is correct. Stage 1 alone, at 54 false alarms/hour with high recall, is the appropriate configuration.
- **As a species-presence record** — asserting that a threatened species was present at a place and time — a confident wrong-species detection enters a conservation database as a false record, which is materially worse than a miss. This is the failure mode species-discrimination AUC measures, and the verified pipeline at 13.4 false alarms/hour is the appropriate configuration.

This paper's contribution targets the second use case. It does not claim superiority for the first.

6. Limitations

Stated plainly, and in order of how much they constrain the conclusions.

1. **No external benchmark validation.** Every comparison here is internal — this pipeline against itself under controlled variation. The DCASE Task 5 development set [1,4] would provide an external, publicly-scored comparison, and the front-end is already configuration-identical to that benchmark's baseline [4], which substantially de-risks the port. It has not been run. It is scheduled after this submission cycle, and its absence is the single largest gap in this work. **No number in this paper should be compared to a published DCASE F-score**, because different data, different task definition, and different metric implementations make such a comparison invalid.
2. **Naturally occurring calls remain unconfirmed.** The background is genuine — every bed is a real Western Ghats soundscape from the same dataset's unlabelled soundscape collection, not silence or synthesised noise. What is planted is the *target calls*, at controlled SNR. So the pipeline has never been asked to find a call that a wild bird actually made, only one inserted into wild audio. This is the most consequential remaining gap after external validation, and unlike field deployment it is not resource-bound: the dataset holds far more unlabelled soundscape than the 14 recordings used here, and scanning it costs roughly one GPU-hour per 33 hours of audio. The obstacle is adjudication, not compute — confirming a detection in unlabelled audio requires someone able to identify the species by ear. §4.12 has now run that scan: 26.6 hours of unlabelled soundscape, 350 detections for the study species against 201 for a control on identical files, and 60 clips saved per species. That changes this limitation from *not attempted* to *attempted and unadjudicated*, which is a smaller change than it sounds. Not one of those detections is confirmed, every score sits within 0.04 of its threshold, and the adjudication step — an expert ear — has still not happened.
3. **The five support calls may characterise a few individuals rather than the species.** Support sets are drawn from the dataset's recordings without regard to where — or from which bird — those recordings were made. *S. albiventris* occupies discontinuous montane cloud forest, distributed from Kodaikanal in the east to Munnar in the west and south to Meghamalai, and the group that collected this dataset's soundscapes has documented cultural and genetic variation across that landscape arising through

separate, asynchronous processes [24]. The obvious worry is therefore dialect: that five calls drawn from one massif describe a population rather than a species. On enquiry, a researcher who works on this species' song gave a more accurate reading — **individual variation is very high, so population-level signatures exist but are not strongly evident** (C. Arvind, pers. comm., 4 Sep 2026). That redirects the limitation rather than dissolving it, and arguably sharpens it. If between-individual variation dominates between-population variation, then five calls — possibly from only one or two birds — are a thin sample of the species along the axis that actually varies most. §4.5 compared four strategies for choosing *which* five calls; it never controlled for how many distinct individuals or recordists they came from. One observation in this paper's own data is consistent with the expert account and is recorded without being leaned on: this species has the lowest mean pairwise call similarity of the ten targets (0.268, against 0.339–0.543 for the rest; `data/argus_species_covariates.csv`). That is a description of a single number and not a predictive claim — the claim that call stereotypy predicts detection difficulty was tested across all ten species and retracted (Appendix A, #43). **The provenance has since been checked**, in response to that exchange, using the `latitude`, `longitude` and `author` fields that were present in the dataset's own `train_metadata.csv` all along and that every run of this pipeline reads and prints without ever having been consulted. The 22 recordings come from **8 distinct recordists**, spanning latitudes 10.08–10.24 N and longitudes 77.07–77.49 E — which, reassuringly, is close to the full east–west extent of the range as described, from the Anamalai end to the Palani Hills. Two recordings carry no coordinates. Geographic coverage is therefore better than feared, and the narrow dialect worry is correspondingly weaker.

The recordist distribution is another matter. **Twelve of the 22 recordings — more than half — come from a single team at a single point** (10.2381 N, 77.4892 E). Drawing five support calls at random from this pool, the expected number originating from that one team is 2.7 of 5; the probability that at least three of the five do is **59%**, and the probability that none do is 1%. Since the expert account holds that individual variation exceeds population-level variation, and since one team recording at one point is the condition most likely to yield repeated recordings of the same few birds, the majority of five-shot draws in this study are likely to have been dominated by a single locality's individuals. That is a concrete, quantified version of the concern, and it is not something any measurement reported in this paper would have detected.

The experiment this now motivates is specific rather than exploratory: stratify the support set by recordist and measure discrimination as a function of how many distinct recordists the five calls span. With 8 recordists and 22 recordings it may still be underpowered, which is worth reporting alongside whatever it finds. A published method for quantifying repertoire complexity in this species specifically also exists [25] and was not used.

4. **Distillation has not removed the Perch dependency, and widening the student does not help.** §4.10 measured how much of stage 2's discrimination survives compressing Perch into the stage-1 network: a mean loss of 0.136 AUC on a 2,636-pair teacher set, heterogeneous across species from near-parity (bkwsti, −0.011) to chance for *S. albiventris* itself (0.511). §4.11 then quadrupled the teacher set and swept the bottleneck from 64 to 512 dimensions. Two things followed. The study species rose off chance to 0.608 at the same width and head, which suggests the first attempt was undersized rather than misconceived — though epochs and batch size changed too, so that is suggestive rather than controlled. And widening the bottleneck did nothing at all: an eightfold widening moved species AUC by 0.0073, inside the noise floor, while fidelity to the teacher rose 0.078. The best distilled arm still trails Perch by 0.094, roughly three times the noise floor. **The pipeline as built still requires Perch at detection time.** What has changed since §4.10 is the explanation, not the conclusion: the gap is not a dimensionality bottleneck, and the remaining candidates — teacher-set scale, per-species coverage, and the distillation objective itself — are untested against one another.
5. **Sample sizes are modest, though less so than they were.** The centrepiece uses 3 encoder seeds per species; its primary paired test carries n = 10 species. The sweeps were widened from 3 species × 2 seeds (6 measurements per condition) to 5 × 3 (15), which clears the 10-per-condition level below which a

difference of this magnitude cannot be separated from noise. That widening changed two results materially — the shot-strategy spread halved to sit exactly on the noise floor, and the decoy-difficulty gain spread halved to sit clearly inside it — in both cases strengthening a conclusion the narrower sampling had reached less securely. Ten *seeds* per condition would be more defensible still; this is a compute-budget constraint (a session is 3–8 GPU-hours on a free tier), stated rather than concealed.

6. **The P2 result rests on consistency more than margin.** The species-matched-negatives finding (§4.9) clears its noise threshold by 0.0006. It is reported as a finding because the direction is negative in 5 of 5 species and because it agrees with an independent ablation result, not because the margin test is comfortable. A wider seed count would strengthen or overturn it.
7. **One species carries the conservation framing, on the thinnest data.** *S. albiventris* motivates the work but is one of ten species tested, and — with 22 recordings against 500 for most others — is the least well-supported target in the set. Its result (0.469 → 0.680) is consistent with the others but rests on the least evidence. Additional recordings of this species do exist in the wider public archive the dataset draws from, so this is a constraint of the chosen dataset rather than an absolute one; expanding it mid-study was rejected because it would break comparability with four archived replications.
8. **Held-out pools are small for low-recordist species.** The recordist-disjoint split (§4.9) necessarily shrinks the held-out pool; for the study species it fell to 17 calls, which is why its bound is wider than others'.
9. **The log-Mel mechanism is characterised but not isolated.** §4.8 identifies *what* goes wrong with a log-Mel front-end in this regime — recall collapses at the proposal stage (−0.192 stage-1 recall) and propagates — which reconciles the divergence from published expectation. What has not been tested is *why* PCEN's per-channel gain control specifically rescues recall under a 250-step fine-tune. A front-end comparison at longer fine-tune budgets would separate "PCEN is better here" from "PCEN adapts faster," and these data cannot distinguish those.

7. Conclusions and Future Work

This work asked whether a bioacoustic detector can learn a new species from five calls and tell it apart from species that sound almost the same.

It can, but not with proposal-only matching. Prototype matching alone discriminates target from nearest confuser at 0.585 mean AUC across ten species — at or below chance for two of them — and additional support examples do not fix this. Adding a verification stage built on pretrained bird-audio embeddings lifts discrimination to 0.751, improving every one of ten species, by a margin 5.2× the empirically measured noise floor, replicated across four independent sessions, while cutting false alarms on target-absent audio fourfold.

Detection metrics and discrimination metrics evaluate distinct failure modes. Soundscape retrieval average precision slightly declines (0.611 → 0.604) while species-discrimination ROC-AUC against acoustic confusers rises from near-chance to 0.751 (0.585 → 0.751, improving 10/10 species). Ranked by soundscape AP, the verified pipeline places last among six candidate fusion rules; ranked by confuser discrimination AUC, it places first. For bioacoustic monitoring where positive detections represent species records rather than generic sound candidates, verifying species identity against near confusers is essential.

Four published techniques did not transfer. A DCASE-2023-winning augmentation, a DCASE-2024-winning front-end, hard-negative mining, and the prototypical probe from Perch 2.0's own benchmark were each tested against the measured noise floor; the untouched baseline beat all four in this five-shot, short-fine-tune regime. Reporting this matters as much as reporting what worked.

Compressing the discriminator into something deployable costs more than it gives back today. Distilling Perch's embeddings into the stage-1 network — so no Perch inference is needed at detection time — was tested directly rather than assumed either way. The measured cost is a 0.136 AUC loss on average, unevenly distributed: near-free for the easiest species (bkwsti, −0.011), chance-level for *S. albiventris* itself (0.511). A follow-up swept the bottleneck width eightfold and found the most interesting of three pre-registered

outcomes: fidelity to the teacher rose steadily (+0.078) while species discrimination did not move at all (0.0073, inside the noise floor). **A student can be trained progressively closer to Perch's embedding geometry and gain nothing at telling similar birds apart** — a caution for anyone compressing a bioacoustic embedding model by matching embeddings and validating on embedding similarity. Quadrupling the distillation set did lift the study species off chance, to 0.608. The architecture is still not reducible to a field-deployable model for the species it was built around, but the reason is not the dimensionality bottleneck it was assumed to be.

Future work, in priority order:

1. **External validation on DCASE Task 5** — the highest-value outstanding item, de-risked by front-end compatibility, requiring only that the correct set (validation, 52.14% F baseline) be named and the within-file adaptation compliance checked.
2. **Test how many distinct individuals the support set spans.** Expert account holds that individual variation in this species' song exceeds population-level variation (§6, limitation 3), which makes recordist- and individual-diversity of the five support calls a more promising axis than geography alone. Measure performance against the number of distinct recordists the shots span, and partition the 22 recordings by coordinates as a secondary test. Placed second because it is nearly free — both fields are already in the dataset metadata — and because it bears on what the headline result means rather than on how far it extends.
3. **A distillation objective that preserves discrimination, not geometry.** §4.11 partially answers what §4.10 left open: more training pairs helped, more bank species helped, a non-linear head did not clear the noise floor, and width did not matter at all. What remains is the sharper question that result raises — cosine similarity to the teacher rose steadily while species discrimination stayed flat, so the objective being optimised is measurably not the objective that matters. A distillation loss defined over species separability rather than embedding proximity is the obvious next experiment, and the frontier data in `data/argus_frontier.csv` gives it a baseline to beat.
4. **Widen seed counts further.** The sweeps now carry 15 measurements per condition and the confounds' unresolved arm was settled at six seeds, but two results remain bounded by their noise rather than by their effect: stage-1's SNR trend (§4.6), where two legitimate tests disagree, and the species-matched-negatives margin (§4.9). Both would be settled by a pre-designed, properly powered test rather than by more data alone.
5. **Field deployment**, on real long-duration recorder data with expert-confirmed labels, converting lab-benchmark evidence into field evidence.
6. **Extend the discrimination metric proposal** into a reusable evaluation protocol other systems could adopt, which is the form in which §4.3 would be most useful to the field.

8. Acknowledgements

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We are grateful to **Chiti Arvind** (Shola Sky Islands Lab, IISER Tirupati), who replied to an unsolicited email from high school students with a correction that changed a stated limitation of this work: that individual variation in *S. albiventris* song exceeds population-level variation, which redirected limitation 3 in §6 from a dialect framing to an individual-sampling one. The distribution of the species described there, and the training-set size of the group's own BirdNET detector cited in §1.1, are also theirs.

We are grateful to **Tom Denton** (Google DeepMind), author of the Perch model this work depends on, who answered a question on the Perch issue tracker about why distillation collapsed unevenly across species, and pointed to the layer-width evidence in [5] — which is why §4.10's follow-up experiment tests the mechanism it does.

Neither of them is responsible for any error here, and neither reviewed this paper.

This work used the publicly available BirdCLEF 2024 dataset [6] and the publicly released Perch v2 model [15]. Computation was performed on a free-tier public GPU platform. Generative AI tools (Claude, Antigravity) wrote much of the code and helped draft the text of this paper. All experiments and results are our own.

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> **Citation note.** Author lists, titles, venues and years above were read from the source PDFs archived in `papers/`, not reproduced from memory or from other papers' bibliographies. Four entries were found misattributed during this pass and corrected: [12] had been credited to the wrong Liu — Lingwen Liu at Jilin Agricultural University, not Haohe Liu at Surrey, a different group entirely; [17], [18] and [19] carried wrong author initials; and [21] had no authors at all. Volume, page and DOI details are given where they were verified and omitted where they were not, rather than being filled in by inference. Two sources, [22] and [23], are public conservation assessments rather than primary literature and are cited as such.

Appendix A — Retraction record

This project maintains a public, dated record of every claim it published and later found to be wrong. A wrong published claim is treated as a more serious failure than a disappointing result. Seven claims carry the strongest verdict:

#	What was claimed	What was actually true
27	"Stage 1 is below chance" as a general claim	True for 2 of 10 species; above chance for the rest
28	"The `rated` shot-selection strategy wins"	Spread (0.068) does not survive a proper two-way test; per-species results disagree; aggregate winner separated by 0.00004 in one run
29	"Performance plateaus at 5 shots"	Only the 1→3 step reliably clears the measured noise floor

#	What was claimed	What was actually true
39	"Stage 1's own specificity degrades with SNR"	Tested against the wrong noise bound — borrowed from the verifier's series, which is 1.7× less noisy. Correctly bounded, the change is unresolved. **Still stands** after re-running at 5 species × 3 seeds: the sweeps' own margin test returns *too close to call*, and although a paired per-species test on the same data is significant (d/SE −4.46, 5/5 species), the retraction is not reversed on the strength of a second, more favourable test. See §4.6
43	"Call stereotypy predicts detection difficulty"	A family-wise permutation test over six screened covariates gives p = 0.253 — appears by chance roughly 1 time in 4
49	The ablation ladder's main effects	Computed with the baseline configuration weighted 3× more than each alternative, inflating every measured deficit by up to 2.4×
50	"The prototypical probe's AUC drop clears the noise floor"	Corrected value falls inside the project's own evidence bar; the rejection survives on false-alarm rate, not AUC

Additionally, four real defects were found and fixed *before* any result was reported: an encoder-bank leak allowing non-primary target species into training data; a held-out pool that leaked support recordings under non-positional shot strategies; a recall-floor threshold that never reached its floor; and a missing Perch peak-normalisation step. All four carry regression tests.

Two of the retractions above (49, 50) were found by an adversarial audit that re-checked every published number against an independent run — and that audit's own recommendations were then re-checked by a further pass, which caught one of its proposed fixes as itself incorrect before it was applied.

Appendix B — Reproducing every number in this paper

All results derive from eleven CSV files in `data/`. No figure in this paper was transcribed from a console log; each was recomputed from the archived raw data during writing.

Section	File	Computation
§4.1, §4.3	`argus_multispecies_results.csv`	Filter `fusion` to `stage-1 only (p)` and `perch only (v)`; group by `species`; compare mean `auc`, `ap`, `f1`
§4.2	same	Same filter; mean `fa_per_hr`
§3.7, §4.8	`argus_ablation_ladder.csv`	Noise floor: `config == "pcen
§4.4	`argus_sweep_shot_count.csv`	Group by `n_shot`
§4.5	`argus_sweep_shot_strategy.csv`	Group by `strategy`
§4.6	`argus_sweep_snr.csv`	Pivot on `snr_db` × `fusion`; note `auc` (all planted) vs `auc_det` (localised only) are different columns answering different questions
§4.7	`argus_sweep_decoy_difficulty.csv`	Group by `decoy_mode`
§4.9	`argus_confounds.csv`	Group by `run_type`, then `arm`
§4.10	`argus_distill.csv`	Group by `backend`; compare mean `auc` for `perch` vs `distilled`, and per-species `auc` for the heterogeneity claim
§4.11	`argus_frontier.csv`	Group by `arm`; `width` against mean `auc` for the flat curve, against `val_cosine` for the rising one. Sort by `width`, not by row order — arms were run in priority order, not width order

Section	File	Computation
§4.12	`argus_wildscan.csv`	Group by `species`; count rows for detections, `nunique` on `file`, and divide by 26.6 h for the rate. `argus_wildscan_clips.npz` holds the 60 top-scoring waveforms per species

A note on §4.6. The distinction between `auc` and `auc_det` is load-bearing and easy to misread. `auc` scores every planted item, assigning 0.0 to any the system never localised; `auc_det` scores only localised items. A summary that quotes one while describing the other will appear contradicted by the raw data. The claim "identification is flat with loudness, detection is not" is true of `auc_det` and coverage respectively, and is false of `auc`.

Source code. Seven files (`arguswala_setup_v2.py`, `arguswala_v2.py`, `arguswala_sweeps.py`, `arguswala_ladder.py`, `arguswala_confounds.py`, `arguswala_distill.py`, `arguswala_frontier.py`), packaged into `ARGUS_Kaggle_Notebook.ipynb` by a generator that verifies every cell's contents against its source file by SHA-256 and independently `ast.parse`s each one. 394 local assertions across five suites (`_test_v2_logic.py`, `_test_ladder.py`, `_test_confounds.py`, `_test_distill.py`, `_test_frontier.py`) run without a GPU, a dataset, or a network connection.

Note: Parts of this document were edited with generative AI assistance (Claude).